Supplementary material for Two-stage MILP scheduling and structure-aware learning-enhanced branch-and-cut for dual-source mixed-flow rotating-chamber cluster tools

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1 Computational evidence design

The computational study separates formulation validation, nominal method comparison, mechanism-specific controls, manufacturing-factor analysis, parameter sensitivity, and distribution-shift evaluation. Table 1 gives the complete prespecified design. A run is identified by the instance, method, solver seed, training seed where applicable, and experimental profile. All observations are retained, including time-limited runs.

Table 1: Prespecified computational suites.

Suite	Design	Runs
Validation	Four small physical cases, one exact solve each	4
Nominal	20 instances; two solver baselines with five solver seeds and five learned configurations with five solver and five training seeds	2700
Two-stage stability	20 instances, three refinement profiles, five solver seeds, and five training seeds	1500
Factorial DOE	$4 \times 5 \times 3$ wafer-count, recipe-composition, and processing-scale cells with five solver seeds	300
SPBS	48 instances, automatic versus no warm start, ten matched solver seeds	960
Sensitivity	Eight one-factor settings with ten solver seeds	80
Distribution shift	Nine out-of-distribution instances; two solver baselines and five learned configurations with matched solver and training realizations	729
Total	Excluding the four validation cases	6269

All nominal, two-stage, factorial, SPBS, and sensitivity runs completed without execution failure. Every saved incumbent used in the reported statistics was checked independently against the corresponding mathematical instance. The distribution-shift suite is analysed using the same retention, validation, and statistical rules after its complete matrix is available.

2 Independent validation and numerical tolerance

The validator reloads each saved solution, fixes the primary decision values, and checks the active physical model independently of the learning policy. The checks cover routing and stage order, chamber and slot exclusivity, robot and load-lock use, recipe grouping, cleaning, residence-time limits, and vacuum-zone dummy-wafer capacity. Continuous residuals use the solver tolerance 10^{-6} plus a 10^{-12} serialization allowance, i.e., an effective threshold of 1.000001×10^{-6}. Binary and integer requirements are not relaxed. This allowance only accounts for the decimal text round trip of an otherwise accepted solver solution.

3 Event-time reconstruction of schedule stability

Schedule-stability metrics are reconstructed from physical event times rather than auxiliary objective variables. This makes the definition identical under all three refinement profiles. For product wafer w, define six nonnegative route waits:

$$q_{w,1} = [s_w^{\mathrm{PM}} - e_w^{\mathrm{VTRload}}]_+, \qquad q_{w,2} = [s_w^{\mathrm{VTRunload}} - e_w^{\mathrm{PM}}]_+, \tag{1}$$
$$q_{w,3} = [s_w^{\mathrm{AL}} - e_w^{\mathrm{ATR,LP\to AL}}]_+, \qquad q_{w,4} = [s_w^{\mathrm{ATR,AL\to LL_u}} - e_w^{\mathrm{AL}}]_+, \tag{2}$$
$$q_{w,5} = [s_w^{\mathrm{VTRload}} - e_w^{\mathrm{LL_u}}]_+, \qquad q_{w,6} = [s_w^{\mathrm{ATR,LL_l\to LP}} - e_w^{\mathrm{LL_l}}]_+, \tag{3}$$

where $[x]_+ = \max(0, x)$ and s and e denote stage start and end times. The route-total and route-maximum metrics are

$$Q_{\mathrm{total}} = \sum_w \sum_{k=1}^{6} q_{w,k}, \qquad Q_{\mathrm{max}} = \max_{w,k} q_{w,k}. \tag{4}$$

Processing cadence is reconstructed from the ordered start times of active full-flow and mixed-flow chamber batches. For adjacent-start intervals $d_1, \ldots, d_m$, cadence variation is $\mathrm{CV}(d) = \mathrm{sd}(d)/\,\mathrm{mean}(d)$. The cadence-deviation sum is the sum of the absolute deviations used by the linear refinement objective after the discrete batch structure has been fixed.

Table 2: Instance-first means in the controlled two-stage experiment.

Profile	Route-total	Route-maximum	Cadence CV	Cadence deviation
Full	2596.01	296.33	**0.130**	**22.04**
Linear-off	**2469.47**	**293.27**	0.134	60.18
Stage2-off	5398.60	952.57	0.216	370.41

Relative to stage2-off, the full profile reduces route-total wait by 51.91%, route-maximum wait by 68.89%, cadence CV by 39.70%, and cadence deviation by 94.05%; all four comparisons remain significant after Holm adjustment. The linear objective also reduces cadence deviation by 63.38% relative to the quadratic profile. Its route-total, route-maximum, and cadence-CV differences from the quadratic profile are not Holm-adjusted significant.

4 Statistical protocol

The manufacturing instance is the unit of inference. Repeated solver and training realizations are first averaged within each instance and method or profile. Mean differences are reported with a nonparametric

95% bootstrap interval. Paired two-sided Wilcoxon signed-rank tests use the same instance pairs, and all planned contrasts within an analysis family are adjusted by the Holm procedure. Failure, incumbent, and optimality indicators are retained as binary outcomes. Primal–dual integral (PDI) is the primary anytime measure; lower values are better. Time-limited runs remain in PDI, time, incumbent, and failure summaries.

5 Independent SPBS warm-start experiment

Table 3 reports the controlled SPBS comparison. Automatic SPBS provides a valid incumbent in every run and reduces PDI by 7194.27 (26.19%). The matched instance-level effects on incumbent availability and PDI both remain significant after multiplicity adjustment.

Table 3: SPBS warm-start comparison over 48 instances and ten matched seeds.

Condition	Runs	Incumbent (%)	Optimal (%)	Mean PDI
Automatic SPBS	480	**100.00**	54.58	**20278.78**
No warm start	480	51.67	51.67	27473.05

The incumbent-rate difference is 48.33 percentage points (95% CI $[34.58, 62.29]$; Holm $p = 2.55 \times 10^{-5}$). The PDI difference is 7194.27 (95% CI $[5145.78, 9317.61]$; Holm $p = 3.46 \times 10^{-8}$). The optimal-rate and elapsed-time differences do not remain significant after Holm adjustment.

6 One-factor parameter sensitivity

Table 4 reports all eight prespecified settings. Each row contains ten default-SCIP runs and every run provides an independently valid incumbent. The experiment evaluates operational solvability under parameter perturbations; because it contains one method, it is not used for a cross- method robustness ranking.

Table 4: One-factor sensitivity results for the 24-wafer 1:1 mixed-flow case.

Setting	Optimal (%)	Mean PDI	Mean time (s)
Nominal	20	40728.91	515.09
Motion scale 0.8	40	37126.93	494.55
Motion scale 1.2	0	42478.32	539.79
Processing scale 0.8	60	33961.82	459.63
Processing scale 1.2	10	38663.67	508.40
Cleaning interval 3	10	38429.95	517.26
Cleaning interval 10	40	28899.41	497.91
Vacuum-zone dummy-wafer inventory 10	20	38152.52	484.80

7 Factorial-model diagnostics

The factorial design contains 60 complete cells and five solver seeds per cell. The prespecified PDI model has $R^2 = 0.9980$. Wafer count and recipe composition exhibit strong interactions, whereas all prespecified processing- scale main-effect and interaction confidence intervals include zero. Figure 1 provides the residual and influence diagnostics.

8 Reproducibility record

The reproducibility package preserves the instance manifest, solver and training seeds, method configuration, time and memory limits, trained-model metadata, raw row records, saved incumbent solutions, independently derived manufacturing metrics, and machine-readable statistical summaries. Every formal suite is accepted only when its expected matrix is complete, duplicate keys and execution errors are absent, saved solutions are covered one-to-one by independent validation, and the statistical input count equals the frozen raw count.

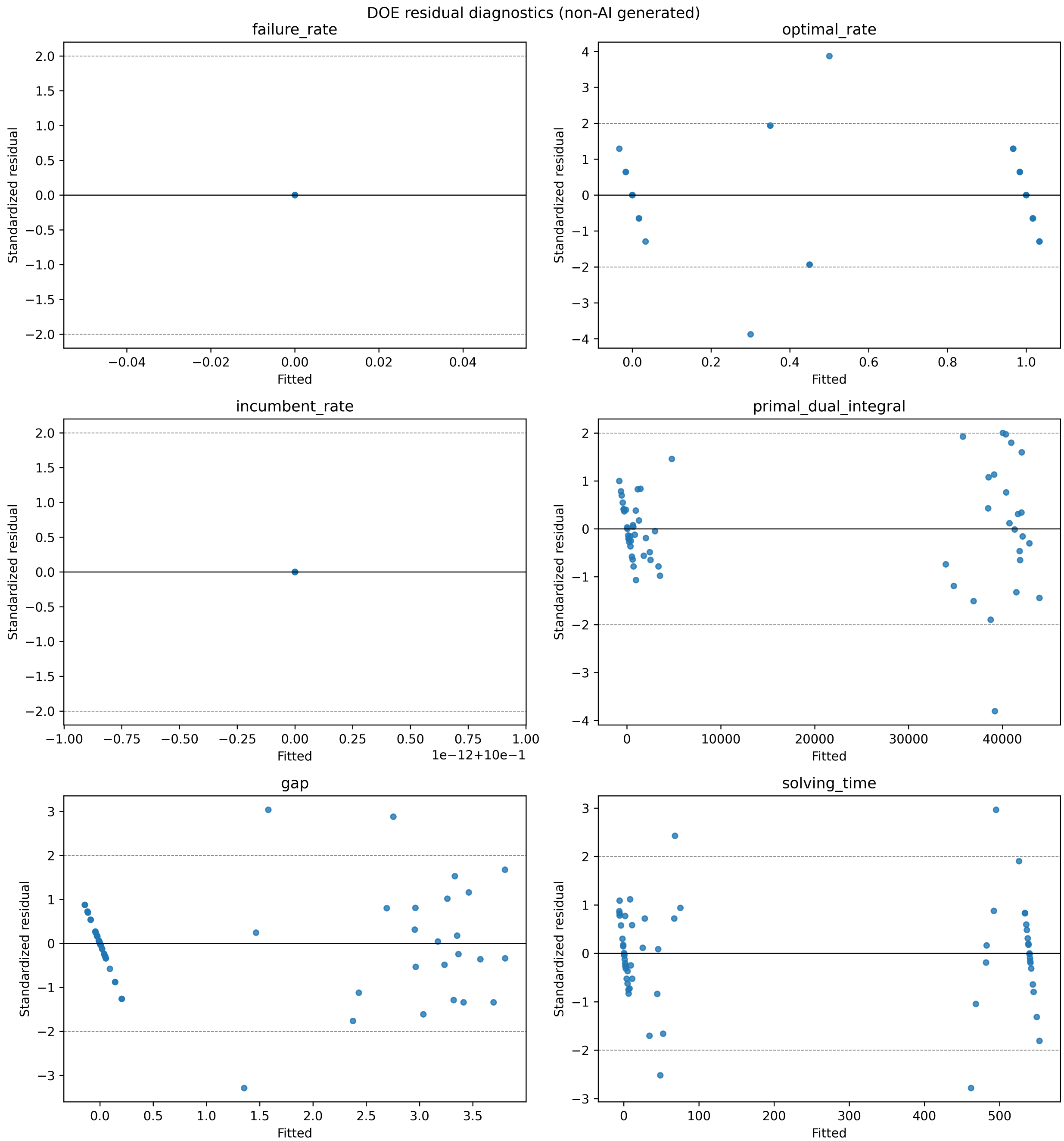

Figure 1: Diagnostic plots for the prespecified factorial PDI model fitted to the complete 300-run design.